

MANISHA KADIRA

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Summary

Aerospace Engineer with strong fundamentals in orbital mechanics, astrodynamics, orbital dynamics, numerical methods, and mission trajectory analysis, built through an M.Tech in Guidance & Control and a B.Tech in Aerospace Engineering, along with hands-on experience using STK for mission analysis. I bring nearly 3 years of experience developing and validating numerical simulation models at Synopsys. I'm looking to bring this combination of core astrodynamics to opportunities across the space domain — orbital safety, mission design, and flight dynamics.

Education

Indian Institute of Science

2021 – 2023

Master of Technology in Aerospace Engineering, (CGPA - 8.8/10)

Bangalore, Karnataka

Relevant Coursework:

Navigation, Guidance and Control

Control Systems Design

Guidance Theory and Applications

Optimization Methods

Motion Planning for Autonomous Systems

Operations Research

Indian Institute of Space Science and Technology

2016 – 2020

Bachelor of Technology in Aerospace Engineering, (CGPA - 7.05/10)

Trivandrum, Kerala

Relevant Coursework:

Automatic Control

Space Flight Mechanics

Atmospheric Flight Mechanics

Modeling and Analysis Lab

Statistics and Probability

Space Mission Design and Optimization

Technical Skills

Tools and Libraries: Python, C++, MATLAB, Simulink, STK

Software Engineering: Docker, Git version control

Experience

Senior Engineer, Applications Engineering

2023 – Present

Ansys, Part of Synopsys

- Develop and validate reduced-order/surrogate models (Ansys Twin Builder, optiSLang, SimAI) for system-level digital twins, and validating to ensure fast models match high-fidelity reference behavior.
- Collaborate with cross-functional team to build system models(Digital Twins) and deliver them as Deployable Containers or Apps.
- Leading a team of 5-members to update content on Ansys Twin Builder to match the release version.
- Delivered multiple workshops, trainings and PoCs to encourage customers to adopt Digital Engineering solutions.

Intern

2022

Ansys, Part of Synopsys

- Worked on creating a scenarios to determine the best possible communication path between a communication tower and a drone in AGI STK software.

Projects

Game Theory based Q Learning for collision avoidance in Multiagent systems | *M. Tech Thesis*

- Developed a Multi-Agent Reinforcement Learning (MARL) based control framework for collision avoidance and coordinated navigation of autonomous agents.
- Implemented Correlated Equilibrium Q-Learning (CE-Q) for action selection under multi-agent interactions.
- Formulated agent interactions as a Markov Game and developed state-action based decision policies for collision-free navigation.

Multiple Delivery Problem using Vector Field Based Guidance | *Course Project*

- Implemented a Vector Field-based guidance law to converge a UAV onto a moving target's trajectory for autonomous rendezvous/delivery, formulating convergence and traverse vector fields from the target's path equation.
- Derived finite-time convergence conditions using Lyapunov stability analysis and validated the guidance law numerically (MATLAB) across multiple scenarios — straight-line interception, constant-distance tracking, and multi-target/circular trajectories.

Motion Planning using Stochastic Multimodal Trajectory Optimization | *Course Project*

- Performed Path planning of Mobile Robot in 2D space while avoiding obstacles, and generated multiple trajectories for Five-link and Six-Link Manipulators between initial and final positions using SMT0.
- Determined multiple modes of the cost function by formulating the trajectory optimization problem as a density estimation problem, and introducing importance sampling based on the cost function.
- Approximated the multi-modal distribution of the trajectories using a Gaussian Mixture Model(GMM). Applied Variational Bayes Estimation Maximization to maximize the log-likelihood for GMM fitting.